Nearest Smaller Values

Time limit: 1.00 s
Memory limit: 512 MB

CSES 1645 | Sorting and Searching | https://cses.fi/problemset/task/1645

Given an array of n integers, your task is to find for each array position the nearest position to its left having a smaller value.

Input

The first input line has an integer n: the size of the array.

The second line has n integers $x_1,x_2,\ldots,x_n$: the array values.

Output

Print n integers: for each array position the nearest position with a smaller value. If there is no such position, print 0.

Constraints

- $1 \le n \le 2 \cdot 10^5$
- $1 \le x_i \le 10^9$

Example

Input
8 2 5 1 4 8 3 2 5

Output
0 1 0 3 4 3 3 7